

Solutions

Numbering follows the lecture notes. For TYPE C exercises the idea is boxed; the rest are routine once the idea is in hand, which is what the types mean.

Lectures 1 and 2

§1.2–§1.4 Definition, existence, independence

1.8 A (a) $\mu \cdot \xi = \xi_1 - \xi_2$ and $\xi^\top \Sigma \xi = 2\xi_1^2 + 2\xi_1\xi_2 + \xi_2^2$, so $\varphi_X(\xi) = \exp(-i(\xi_1 - \xi_2) - \frac{1}{2}(2\xi_1^2 + 2\xi_1\xi_2 + \xi_2^2))$. Proposition 1.4 gives $\partial_1 \partial_2 \varphi_X(0) = -\Sigma_{12} - \mu_1 \mu_2 = -1 + 1 = 0 = -\mathbb{E}[X_1 X_2]$, so $\mathbb{E}[X_1 X_2] = 0$ — uncorrelated only after the means are subtracted.

(b) With $B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, Theorem 1.5 gives $BX \sim \mathcal{N}((0, 2), B\Sigma B^\top)$ and $B\Sigma B^\top = \begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix}$. So $X_1 + X_2 \sim \mathcal{N}(0, 5)$, $X_1 - X_2 \sim \mathcal{N}(2, 1)$, and their covariance is $1 \neq 0$: not independent.

(c) $\text{Cov}(X_1 + X_2, X_1 - X_2) = \Sigma_{11} - \Sigma_{22}$, and the pair is jointly Gaussian, so Theorem 1.16 makes independence equivalent to $\Sigma_{11} = \Sigma_{22}$ — equal variances, whatever the correlation. And $v \cdot X$ is constant iff $\text{Var}(v \cdot X) = v^\top \Sigma v = 0$, i.e. iff $v \in \ker \Sigma$, so such a v exists exactly when Σ is singular.

1.9 C

Apply the invariance to the **transform**, and rotate $(\xi_1, \xi_2, 0, \dots, 0)$ onto $(\sqrt{\xi_1^2 + \xi_2^2}, 0, \dots, 0)$. Independence makes φ_X a product; the rotation turns that product into Cauchy's functional equation in ξ^2 .

Independence gives $\varphi_X(\xi) = \prod_j \varphi_1(\xi_j)$, where φ_1 is the common transform of the coordinates. Invariance gives $\varphi_X(O^\top \xi) = \varphi_{OX}(\xi) = \varphi_X(\xi)$ for all O , so φ_X is a function of $|\xi|$ alone. Taking $O = \text{diag}(-1, 1, \dots, 1)$ gives $X_1 \stackrel{d}{=} -X_1$, so φ_1 is real and even.

Now take $\xi = (\xi_1, \xi_2, 0, \dots, 0)$ and an O carrying it to $(r, 0, \dots, 0)$ with $r = \sqrt{\xi_1^2 + \xi_2^2}$. Since $\varphi_1(0) = 1$, the two products agree:

$$\varphi_1(\xi_1) \varphi_1(\xi_2) = \varphi_1(\sqrt{\xi_1^2 + \xi_2^2}).$$

Set $F(u) = \varphi_1(\sqrt{u})$ for $u \geq 0$. Then $F(u)F(v) = F(u+v)$, $F(0) = 1$, and F is continuous. Also $F(u) = F(u/2)^2 \geq 0$; if $F(u_0) = 0$ for some u_0 then $F(u_0 2^{-k}) = 0$ for every k , contradicting $F(0) = 1$ and continuity. So $F > 0$, $\log F$ is continuous and additive, hence linear, and $F(u) = e^{-cu/2}$. Finally $|F| \leq 1$ forces $c \geq 0$. Therefore $\varphi_1(\xi) = e^{-c\xi^2/2}$ and $\varphi_X(\xi) = e^{-c|\xi|^2/2}$, i.e. $X \sim \mathcal{N}(0, cI_d)$.

No moment hypothesis was used anywhere; the variance appears at the end as the constant c . Note also where $d \geq 2$ entered: with one coordinate there is nothing to rotate, and $O(1) = \{\pm 1\}$ says only that X_1 is symmetric.

1.14 A $\hat{f}(\xi) = \int f(x) e^{-ix \cdot \xi} dx = \mathbb{E}[e^{-iZ \cdot \xi}]$ for $Z \sim \mathcal{N}(0, I_d)$, which factors into d copies of Lemma 1.10: $\hat{f}(\xi) = \prod_j e^{-\xi_j^2/2} = e^{-|\xi|^2/2}$.

For the inversion, each coordinate integral is $\int_{\mathbb{R}} e^{-\xi^2/2} e^{ix\xi} d\xi = \sqrt{2\pi} \mathbb{E}[e^{-i(-x)g}] = \sqrt{2\pi} e^{-x^2/2}$, again Lemma 1.10. Multiplying d of them and dividing by $(2\pi)^d$ gives $(2\pi)^{-d/2} e^{-|x|^2/2} = f(x)$. Since $\hat{f} = (2\pi)^{d/2} f$, the standard Gaussian density is an eigenfunction of the transform.

1.15 B (a) $Z = A^{-1}(X - \mu)$, so $p_X(x) = p_Z(A^{-1}(x - \mu)) |\det A|^{-1}$. Now $|A^{-1}(x - \mu)|^2 = (x - \mu)^\top A^{-\top} A^{-1} (x - \mu) = (x - \mu)^\top \Sigma^{-1} (x - \mu)$ because $\Sigma^{-1} = (AA^\top)^{-1} = A^{-\top} A^{-1}$, and $|\det A| = (\det \Sigma)^{1/2}$. Both quantities are functions of Σ alone, so the formula is independent of the square root chosen; concretely, any two differ by $A' = AO$ with O orthogonal, and $|\det A'| = |\det A|$.

(b) If Σ is singular pick $v \in \ker \Sigma$, $v \neq 0$. Then $\text{Var}(v \cdot X) = v^\top \Sigma v = 0$, so $v \cdot X = v \cdot \mu$ and X lies in the hyperplane $\{x : v \cdot x = v \cdot \mu\}$. Integrating any density over a hyperplane gives 0, so X has none.

1.18 B (a) $X + Y = (1 + \varepsilon)g$, which is 0 when $\varepsilon = -1$ and $2g$ when $\varepsilon = +1$: a half-and-half mixture of δ_0 and $\mathcal{N}(0, 4)$. It has an atom of mass $\frac{1}{2}$ at the origin. A Gaussian law is either a point mass or has a density, and this is neither.

(b) $sX + uY = (s + u\varepsilon)g$, a half-and-half mixture of $\mathcal{N}(0, (s+u)^2)$ and $\mathcal{N}(0, (s-u)^2)$, with transform $\frac{1}{2}e^{-(s+u)^2/2} + \frac{1}{2}e^{-(s-u)^2/2}$ at $\xi = (s, u)$. That is Gaussian only when $(s+u)^2 = (s-u)^2$, i.e. $su = 0$. So the hypothesis of Corollary 1.7 holds *only* on the two axes, which is exactly the statement that the marginals are Gaussian; off the axes it fails. The corollary is not contradicted — it never applied.

(c) No: (X, Y) lies on $\{y = x\} \cup \{y = -x\}$, and integrating a density over two lines gives 0.

§1.5* Entropy, and why the Gaussian maximizes it

1.19 A (a) $H(q) = -q \log q - (1-q) \log(1-q)$: symmetric about $q = \frac{1}{2}$, zero at the endpoints, maximal at $\frac{1}{2}$ with value $\log 2$. Uniform on n points: $\log n$.

(b) $\log a$; $1 - \log \lambda$; $\frac{1}{2} \log(2\pi e \sigma^2)$. Uniform on $[0, a]$ with $a < 1$ has $h = \log a < 0$. Discrete entropy cannot be negative because $-p \log p \geq 0$ for $p \in (0, 1]$; a *density* may exceed 1, which is exactly what breaks that.

(c) $p_Y(y) = |a|^{-1} p_X(y/a)$; substituting $x = y/a$, $h(Y) = -\int p_X(x) [\log p_X(x) - \log |a|] dx = h(X) + \log |a|$.

1.20 B $-D(p \| q) = \int p \log \frac{q}{p} \leq \log \int q = 0$ by Jensen, with equality iff q/p is constant, hence $\equiv 1$. Taking $q \equiv 1/n$ uniform on n points, $0 \leq D(p \| q) = -H(p) + \log n$, with equality only for $p = q$. Under $X \mapsto aX$ both densities gain the same factor $|a|^{-1}$ and the same substitution is made, so $D(p \| q)$ is unchanged.

1.21 B $\log q(x) = -\frac{d}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma - \frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)$ is quadratic in x , so $\int p \log q$ is a fixed linear functional of $\int p$, $\int x_i p$, $\int x_i x_j p$. Matching mean and covariance therefore gives $\int p \log q = \int q \log q = -h(q)$, and $0 \leq D(p \| q) = -h(p) + h(q)$. For $h(q)$: $\mathbb{E}[(X - \mu)^\top \Sigma^{-1} (X - \mu)] = \text{tr}(\Sigma^{-1} \Sigma) = d$, so $h(q) = \frac{1}{2} \log((2\pi e)^d \det \Sigma)$.

1.22 C

$(x, y) \mapsto (\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}})$ is **orthogonal**: Jacobian 1, so it preserves joint differential entropy.

With $U = \frac{X+Y}{\sqrt{2}}$, $V = \frac{X-Y}{\sqrt{2}}$: $h(U, V) = h(X, Y) = 2h(X)$ by independence. Subadditivity $h(U, V) \leq h(U) + h(V)$ is Exercise 1.20 applied to the joint density against the product of its marginals. Symmetry gives $(X, Y) \stackrel{d}{=} (X, -Y)$, so $V \stackrel{d}{=} U$. Chaining, $2h(X) \leq 2h(U)$.

1.23 C

The Gaussian proof never used Gaussianity. It used that $\log q$ is an **affine combination of the constrained functions**, so $\int p \log q$ is fixed by the constraints alone.

$\log q = \sum_m \lambda_m T_m - \log Z$, so for any feasible p , $0 \leq D(p \| q) = -h(p) - \sum_m \lambda_m c_m + \log Z$ — and the last two terms do not depend on p . Taking $p = q$ evaluates them as $h(q)$; subtracting gives $h(p) \leq h(q)$, equality iff $p = q$. (No integrability of $h(p)$ need be assumed: $-\infty$ makes it trivial, $+\infty$ is excluded by $D(p \| q) \geq 0$.)

Instances: no constraint on $[0, a]$ gives q constant; $T_1(x) = x$ on $[0, \infty)$ gives the exponential; T 's $= x_i, x_i x_j$ give $\mathcal{N}(\mu, \Sigma)$.

Why the assumption is not free. On $S = \mathbb{R}$ with only the mean constrained, $e^{\lambda x}$ is never integrable, and correctly so: $\mathcal{N}(0, \sigma^2)$ is feasible for every σ with $h = \frac{1}{2} \log(2\pi e \sigma^2) \rightarrow \infty$. “ **q is a density**” is exactly what rules this out, and a student who assumes existence has assumed the content.

§2.3 The tower property

2.6 A Corollary 2.5 reads $\text{Var } X = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y])$ and the first term is non-negative. Equality holds exactly when $\text{Var}(X | Y) = 0$, i.e. when X is a function of Y . All the variance sits in the first term exactly when $\mathbb{E}[X | Y]$ is constant, as happens when $X \perp\!\!\!\perp Y$.

2.7 B Given J , X has mean ± 1 and variance σ^2 , so $\text{Var}(X | J) = \sigma^2$ identically and $\mathbb{E}[\text{Var}(X | J)] = \sigma^2$, while $\mathbb{E}[X | J] = \pm 1$ with $\text{Var}(\mathbb{E}[X | J]) = 1$. The total is $\sigma^2 + 1$. Directly: $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = \mathbb{E}[\mathbb{E}[X^2 | J]] = \mathbb{E}[\sigma^2 + 1] = \sigma^2 + 1$.

For the second part, W is a function of $X_1, \dots, X_n$, so conditioning on the X_i is the finer conditioning and Proposition 2.4 gives

$$\mathbb{E}[V | W] = \mathbb{E}[\mathbb{E}[V | X_1, \dots, X_n] | W] = \mathbb{E}[-W/n | W] = -W/n,$$

the last step because a function of W is its own conditional expectation given W . The property is the **tower property**.

§2.5 Bayesian linear models, §2.6* and §2.7*

2.12 A (a) $p(y | w) = (2\pi\sigma^2)^{-n/2} e^{-\|y - Hw\|^2 / 2\sigma^2}$, so $-\log p = \frac{1}{2\sigma^2} \|y - Hw\|^2 + \frac{n}{2} \log(2\pi\sigma^2)$ and the minimizer over w is the least-squares fit. Choosing squared error is asserting that the noise is Gaussian, homoscedastic and independent across observations; most people choosing it are not asserting anything, which is the point.

(b) The posterior density is $\propto \exp\left(-\frac{1}{2\sigma^2}\|y - Hw\|^2 - \frac{1}{2\tau^2}\|w\|^2\right)$, so its maximizer minimizes $\|y - Hw\|^2 + \lambda\|w\|^2$ with $\lambda = \sigma^2/\tau^2$. Large λ means noisy data against a confident prior, so heavy shrinkage; $\tau \rightarrow 0$ gives $\hat{w} = 0$ and $\tau \rightarrow \infty$ gives least squares.

2.13 B (w, y) is a linear image of (w, ε) , hence jointly Gaussian, with $\Sigma_{ww} = \tau^2 I_p$, $\Sigma_{wy} = \mathbb{E}[w(Hw + \sigma\varepsilon)^\top] = \tau^2 H^\top$, $\Sigma_{yy} = \tau^2 HH^\top + \sigma^2 I_n$. Theorem 2.10 then gives the two displays directly. $\text{Cov}(w | y)$ contains no y : the data changes your *estimate*, never your *uncertainty*. Operationally, a surprising y moves the posterior mean and leaves the error bars exactly where they were — special to the Gaussian model, and false in general.

2.14 B $\text{Cov}(X, Y) = \mathbb{E}[(Y^2 - 1)Y] = \mathbb{E}Y^3 = 0$, so the best linear predictor is 0 with error $\text{Var } X = \text{Var}(Y^2) = 2$. Here X is itself a function of Y , so $\mathbb{E}[X | Y] = Y^2 - 1 = X$ with error 0. What fails is **joint** Gaussianity of (X, Y) — X is not Gaussian at all — so Theorem 2.9 does not apply, and the conditional mean is not linear.

2.15 C

$H^\top(HH^\top + \lambda I_n) = (H^\top H + \lambda I_p)H^\top$ — one line of expansion. Multiply by the two inverses.

Both sides equal $H^\top HH^\top + \lambda H^\top$. Left-multiplying by $(H^\top H + \lambda I_p)^{-1}$ and right-multiplying by $(HH^\top + \lambda I_n)^{-1}$ gives the identity. The right-hand form inverts an $n \times n$ matrix, so the cost is $O(n^3)$; this is the kernel trick, and why Gaussian process regression is feasible.

As $\tau \rightarrow \infty$ we have $\lambda \rightarrow 0$, and in the form $H^\top(HH^\top + \lambda I_n)^{-1}y$ the limit is benign because $HH^\top$ is $n \times n$ and invertible: $\mathbb{E}[w | y] \rightarrow H^\top(HH^\top)^{-1}y$, the minimum-norm solution of $Hw = y$. In the other form the limit is $0 \cdot \infty$ — which is what makes this a C and not a computation.

2.17 A (X, ξ) is jointly Gaussian, so $Y = X + \sigma\xi \sim \mathcal{N}(0, (\tau^2 + \sigma^2)I_d)$ by Theorem 1.5 of Lectures 1–2. Hence $\log p_Y(y) = c - \frac{|y|^2}{2(\tau^2 + \sigma^2)}$ and $\nabla \log p_Y(y) = -\frac{y}{\tau^2 + \sigma^2}$, so Proposition 2.16 gives

$$\mathbb{E}[X | Y = y] = y - \frac{\sigma^2}{\tau^2 + \sigma^2} y = \frac{\tau^2}{\tau^2 + \sigma^2} y.$$

Theorem 2.9 gives $\Sigma_{XY}\Sigma_{YY}^{-1}y = \tau^2 I((\tau^2 + \sigma^2)I)^{-1}y$, the same. Only the second derivation used that X is Gaussian; in the first, Gaussianity of X entered nowhere except to make p_Y available in closed form.

2.18 B $p_Y(y) = \mathbb{E}[\phi_\sigma(y - X)]$. Every derivative of ϕ_σ is bounded and, on a compact set of y , bounded uniformly in X ; dominated convergence therefore justifies differentiating under the expectation to all orders, so $p_Y \in C^\infty$ with $\nabla p_Y(y) = \mathbb{E}[\nabla \phi_\sigma(y - X)]$.

For the second integration by parts, $\int \nabla f p_Y = -\int f \nabla p_Y$ requires $f p_Y \rightarrow 0$ at infinity. Taking f smooth with compact support makes the boundary term vanish outright, and that suffices: such f are dense in $L^2(p_Y)$, so orthogonality against them is orthogonality against all competitors in Lemma 2.3(ii). *Both steps are free because Y is a convolution with a Gaussian; an arbitrary Y need not have a density, let alone a differentiable one.*

2.19 C

Tweedie's formula **is** the dictionary. It is an explicit pointwise bijection between the optimal denoiser and the score, so producing either produces the other.

By Definition 2.2 the least-squares minimizer over all D is $D(y) = \mathbb{E}[X | Y = y]$, and Proposition 2.16 says $D(y) = y + \sigma^2 \nabla \log p_Y(y)$. Hence

$$\nabla \log p_Y(y) = \frac{D(y) - y}{\sigma^2},$$

and conversely. Neither direction mentions p_X .

Fitting D requires only pairs $(X, X + \sigma\xi)$, which can be generated from samples of X by adding noise; neither p_X nor a normalizing constant appears. Module II.11 uses the score at every noise level σ to run the noising backwards.